

Quantitative Aptitude

Missing Number Series

Level-1

Q1 Directions: Find the missing term (?) in the given series.

140 , 169 , ? , 257 , 316 , 385 .

- (A) 207 (B) 208
(C) 209 (D) 210
(E) None of these

Q2 Find the missing term in the given series

17, 21, 26, 32, 39, 47, _____, 66

- (A) 55 (B) 56
(C) 57 (D) 58
(E) None of these

Q3 What will come in place of ' ? ' in the given question?

58, 138, 178, 198, 208, ?

- (A) 228 (B) 213
(C) 218 (D) 248
(E) 259

Q4 What will come in place of question mark in the following number series?

9, 11, 22, 51, 107, ?

- (A) 225 (B) 154
(C) 125 (D) 199
(E) 210

Q5 What will come in place of question mark in the following number series?

112, 225, 345, 479, 634, ?

- (A) 854 (B) 817
(C) 652 (D) 859
(E) 742

Q6 Find the missing term.

656 , 432 , 320 , 264 , 236 , ?

- (A) 230 (B) 223
(C) 232 (D) 229
(E) 222

Q7 Find the missing term.

10080 , 1440 , 240 , ? , 12 , 4

- (A) None of these (B) 52
(C) 48 (D) 40
(E) 46

Q8 Find the missing term.

10, 12, 15, 20, 27, 38, ?

- (A) None of these (B) 51
(C) 49 (D) 55
(E) 53

Q9 Find the missing term.

12 , 19 , 35 , 59 , 90 , ?

- (A) 112 (B) 125
(C) 127 (D) 100
(E) 99

Q10 Find the missing term.

37 , ? , 103 , 169 , 257 , 367

- (A) 71 (B) 55
(C) 67 (D) 59
(E) 61

Q11 Find the missing term.

29 , 31 , 37 , 49 , 69 , ?

- (A) 88 (B) 103
(C) 94 (D) 99
(E) 108

Q12 Find the missing term.

125, 80, 45, ?, 5

- (A) 50 (B) 20
(C) 25 (D) 17
(E) 19

Q13 Find the missing term.

0 , 5 , 18 , 43 , 84 , 145 , ?

- (A) 270 (B) 280



- (C) 260 (D) 240
(E) 230
- Q14** Find the missing term.
62, 87, 187, 412, 812, ?
(A) 1458 (B) 1457
(C) 1337 (D) 1437
(E) 1012
- Q15** What will come in place of question marks in the following number series?
12, 12, 18, 36, 90, 270, ?
(A) 945 (B) 814
(C) 652 (D) 859
(E) 800
- Q16** Directions: Find the missing term (?) in the given series.
185, 266, 330, 379, 415, ? .
(A) 440 (B) 435
(C) 450 (D) 455
(E) None of these
- Q17** Directions: Find the missing number in the following number series.
- 3, 83, 152, 208, 249, ?
(A) 280 (B) 320
(C) 265 (D) 351
(E) 273
- Q18** Directions: What will come in place of the question mark (?) in the following questions?
15, 28, 48, 82, 137, ?
(A) **210** (B) **220**
(C) **230** (D) **200**
(E) None of these
- Q19** Find the wrong number in the following number series –
21, 95, 167, 235, 300, 351, 395
(A) 21 (B) 235
(C) 351 (D) 300
(E) None of these
- Q20** Directions: What will come in place of the question mark (?) in the following questions?
9600, 1920, 480, ?, 80, 80
(A) 200 (B) 320
(C) 80 (D) 240
(E) 160



Level-2

- Q1** Find the missing term.
960 , 839 , 758 , 709 , ? , 675
(A) 684 (B) 678
(C) 688 (D) 700
(E) 696
- Q2** *What will come in place of ' ? ' in the given question?*
5, 10, 17, 26, 37, ?
(A) 48 (B) 50
(C) 46 (D) 49
(E) 67
- Q3** In each of these questions a number is missing in the series. Find out the missing number.
10, 20, 22, ? , 46, 92, 94
(A) 44 (B) 24
(C) 40 (D) 26
(E) 48
- Q4** In each of these questions a number is missing in the series. Find out the missing number.
18, 127, 38, ? , 66 , 731
(A) 375 (B) 365
(C) 345 (D) 355
(E) None of these
- Q5** In each of these questions a number is missing in the series. Find out the missing number.
181, 361, 720, ? , 2870, 5735
(A) 1347 (B) 1473
(C) 1437 (D) 1374
(E) None of these
- Q6** Find the missing term.
7 , 16 , 45 , 184 , 915 , ?
(A) 5496 (B) 5444
(C) 5454 (D) 5555
(E) 5478
- Q7** Find the missing term.
15 , 21 , 38 , 65 , 101 , ?
(A) 145 (B) 130
(C) 150 (D) 141
(E) 123
- Q8** Find the missing term.
0 , 2 , 8 , 14 , ? , 34 , 48
(A) 24 (B) 25
(C) 26 (D) 23
(E) 22
- Q9** Find the missing term.
12 , 14 , 32 , 102 , 416 , 2090 , ?
(A) 12500 (B) 11600
(C) 12552 (D) 13458
(E) 11554
- Q10** Find the missing term.
14 , 6 , 5 , 6.5 , 12 , ?
(A) 35 (B) 33
(C) 23 (D) 27
(E) 29
- Q11** Find the missing term.
17 , 16 , 30 , 87 , 344 , ?
(A) 1715 (B) 1685
(C) 1660 (D) 1760
(E) 1735
- Q12** Find the missing term.
24 , 28 , 19 , 35 , 10 , ?
(A) 15 (B) 46
(C) 16 (D) 36
(E) 26
- Q13** Find the missing term.
26 , 12 , 11 , 15.5 , 30 , ?
(A) 78 (B) 82
(C) 74 (D) 68
(E) 72
- Q14** Find the missing term.
3 , ? , 14 , 55 , 274 , 1643
(A) 7 (B) 8
(C) 6 (D) 5



(E) 11

Q15 Find the missing term.

39, 52, 78, 117, 169, ?

- (A) 255 (B) 256
(C) 234 (D) 182
(E) 246

Q16 A series 29, 31, 41, 71, 139, 269

If another series __, __, 84, __, m follows the same pattern as the given number series, then find the value of m.

- (A) 182 (B) 132
(C) 172 (D) 194
(E) None of these

Q17 A series is 8, 7, 10, 21, 68, 315.

If another series 12, __, __, __, __, m; follows the same pattern as the given number series, then find the value of m.

- (A) 878 (B) 600
(C) 546 (D) 795
(E) None of these

Q18 36, 19, 21, 45, Y, 1477

If $A + 124 = 2Y - 48$, then which of the following will be correct?

- (A) A is a perfect square
(B) A is an odd number
(C) A is a prime number
(D) Sum of unit digit and hundred placed digit of A is a prime number
(E) Both (a) and (d)

Q19 9050, 5675, x, 2147, y, 1075, 950

Calculate values of 'x' and 'y', then answer the question that follows. double the value of x let it be 'a'. Then find the sum of a and 6th Term of the given series.

- (A) 8033 (B) 1097
(C) 2111 (D) 3521
(E) 2025

Q20 What will come in place of question mark in the following number series?

29, 33, 60, 76, x, y

Find $(x - y)^2$

- (A) 1254 (B) 1356
(C) 1296 (D) 1547
(E) 1000



Level-3

Directions (1-2) Read the following passage and answer the given questions.

Directions : Study the following information carefully and answer the questions given beside.

Solve both the series given below and answer the following question.

Series I: a, 182, 132, 56, 30, 12, b

Series II: a + c, a, a - c, a - 2c, a - 3c, b, d

Q1 What is the value of $\frac{a}{c}$?

- (A) 2.96 (B) 3.48
(C) 3.78 (D) 4.08
(E) None of these

Q2 What is the value of 'b - d'?

- (A) $\frac{a}{4}$ (B) b
(C) c (D) 2d
(E) a-d

Directions (3-4) Read the following passage and answer the given questions.

Directions : Study the following information carefully and answer the questions given beside.

Both the series given below follow a similar pattern. Solve both the series and answer the following questions.

I: a, 154, 1548, 12390, 74344, 297378

II: b, c, 490, 4908, d, 235624

Q3 What is the value of " $\frac{d}{a}$ "?

- (A) 3225.50 (B) 3272.50
(C) 3333.33 (D) 3600
(E) 3728.50

Q4 The series given below follows a particular pattern. What is the value of 'e'?

in the following series?

$$\frac{a}{3}, 4b, \frac{c}{2.50}, 4b^3, e$$

- (A) 81 (B) 63
(C) 64 (D) 82
(E) 125

Q5 A series is given below, in which P and Q are missing terms.

14, P, 62, Q, 254

If the ratio of P to Q is 5: 21 and Q is 1 more than the cube of 3rd least prime number, then find the next term of the series?

- (A) 508 (B) 500
(C) 510 (D) 504
(E) 520

Q6 A series is given below with one wrong term P.

8, 20, 150, 750, 7500, 93750

A new series is formed, whose 1st term is P followed by (1st term + 82), (2nd term - 72), (3rd term + 62) and (4th term - 52). What will be the 5th term of the series?

- (A) 776 (B) 34
(C) 93776 (D) 176
(E) 7526

Q7 A series is given below, in which X and Y are missing terms of the series.

18, 37, 75, 151, X, Y

A: Both X and Y are multiple of 3.

$$B: X = \frac{Y-1}{2}$$

C: Y is a prime number.

- (A) Only B and C (B) All A, B and C
(C) Only B (D) Only A and C
(E) Only C

Q8 Direction: Study the data carefully and answer the following question.

A series is given below with missing term P.

325, P, 136, 109, 101, 100



A new series is formed, whose 1st term is P and follows the same pattern as in the given series. Find the 4th term of the new series?

- (A) -49 (B) -16
(C) -25 (D) -9
(E) 16

Q9 Direction: Study the data carefully and answer the following question.

A series is given below with missing term P.

8, P, 20, 63, 256, 1285

A new series is formed, whose 1st term is $(P - 3)$ and follows the same pattern. If its n^{th} term is 208, then find the value of 'n'?

- (A) 6th term (B) 5th term
(C) None of these (D) 3rd term
(E) 7th term

Q10 Direction: Study the data carefully and answer the following question.

A series is given below, with one missing term.

8, 4, 4, 8, ?, 256

Another series is formed, whose 1st term is the missing term of the given series, followed by (1st term $\times 0.25$), (2nd term $\times 0.5$), (3rd term $\times 1$), (4th term $\times 2$) and so on. What is the 6th term of the new series?

- (A) 64 (B) 8
(C) 32 (D) 16
(E) 24



Answer Key

Level-1

Q1 (B)
Q2 (B)
Q3 (B)
Q4 (D)
Q5 (B)
Q6 (E)
Q7 (C)
Q8 (B)
Q9 (C)
Q10 (D)

Q11 (D)
Q12 (B)
Q13 (E)
Q14 (D)
Q15 (A)
Q16 (A)
Q17 (E)
Q18 (B)
Q19 (D)
Q20 (E)



Level-2

Q1 (A)

Q2 (B)

Q3 (A)

Q4 (C)

Q5 (C)

Q6 (A)

Q7 (A)

Q8 (A)

Q9 (C)

Q10 (E)

Q11 (A)

Q12 (B)

Q13 (C)

Q14 (D)

Q15 (C)

Q16 (A)

Q17 (D)

Q18 (E)

Q19 (A)

Q20 (C)



Level-3

Q1 (D)

Q2 (C)

Q3 (B)

Q4 (C)

Q5 (C)

Q6 (D)

Q7 (A)

Q8 (B)

Q9 (B)

Q10 (C)



Hints & Solutions

Level-1

Q1 Text Solution:

140, 169, 208, 257, 316, 385

29 39 49 59 69

? = 208

Q2 Text Solution:

The given series is

17, 21, 26, 32, 39, 47, _____, 66

This follows the pattern:


Q3 Text Solution:

Here the pattern is,

$$138 = 58 + 80$$

$$178 = 138 + 40$$

$$198 = 178 + 20$$

$$208 = 198 + 10$$

The next term will be $208 + 5 = 213$

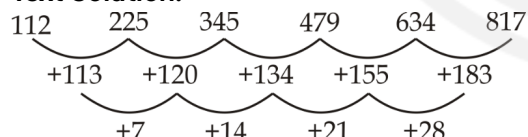
Hence, Option B is Correct

Q4 Text Solution:

9 11 22 51 107 199

2 11 29 56 92

9 18 27 36

Q5 Text Solution:


Ans = 817

Q6 Text Solution:

Pattern of the series is as follows :

$$656 - 224 = 432$$

$$432 - 112 = 320$$

$$320 - 56 = 264$$

$$264 - 28 = 236$$

$$236 - 14 = \mathbf{222}$$

Q7 Text Solution:

The pattern of the series is as follows :

$$10080 \div 7 = 1440$$

$$1440 \div 6 = 240$$

$$240 \div 5 = 48$$

$$48 \div 4 = 12$$

$$12 \div 3 = 4$$

Hence, the answer is 48.

Q8 Text Solution:

Pattern of the series is as follows:

$$10 + 2 = 12$$

$$12 + 3 = 15$$

$$15 + 5 = 20$$

$$20 + 7 = 27$$

$$27 + 11 = 38$$

$$38 + 13 = 51$$

Hence, the answer is 51.

Q9 Text Solution:

Pattern of the series is as follows :

$$12 + 7 = 19$$

$$19 + 16 = 35$$

$$35 + 24 = 59$$

$$59 + 31 = 90$$

$$90 + 37 = 127$$

$$\text{Here, } 7 + 9 = 16$$

$$16 + 8 = 24$$

$$24 + 7 = 31$$

$$31 + 6 = 37$$

$\therefore$ The answer is 127.

Q10 Text Solution:

Pattern of the series is as follows :

$$37 + 22 \times 1 = \mathbf{59}$$

$$59 + 22 \times 2 = 103$$

$$103 + 22 \times 3 = 169$$

$$169 + 22 \times 4 = 257$$

$$257 + 22 \times 5 = 367$$

$$? = 59$$

$\therefore$ The answer is 59.

Q11 Text Solution:

Pattern of the series is as follows :

$$29 + 2 \times 1 = 31$$

$$31 + 3 \times 2 = 37$$

$$37 + 4 \times 3 = 49$$



$$49 + 5 \times 4 = 69$$

$$69 + 6 \times 5 = 99$$

∴ The answer is 99.

Q12 Text Solution:

Pattern of the series is as follows:

$$125 \quad 80 \quad 45 \quad \mathbf{20} \quad 5$$

$$-45 \quad -35 \quad -25 \quad -15$$

$$+10 \quad +10 \quad +10$$

$$? = 20$$

∴ The answer is 20.

Q13 Text Solution:

Pattern of the series is as follows :

$$0 \quad 5 \quad 18 \quad 43 \quad 84 \quad 145 \quad 230$$

$$+5 \quad +13 \quad +25 \quad +41 \quad +61 \quad +85$$

$$+8 \quad +12 \quad +16 \quad +20 \quad +24$$

$$+4 \quad +4 \quad +4 \quad +4$$

$$? = 230$$

∴ The answer is 230.

Q14 Text Solution:

Pattern of the series is as follows :

$$62 + (5)^2 = 87$$

$$87 + (10)^2 = 187$$

$$187 + (15)^2 = 412$$

$$412 + (20)^2 = 812$$

$$812 + (25)^2 = 1437$$

Q15 Text Solution:

The given series is

$$12 \times 1 = 12,$$

$$12 \times 1.5 = 18,$$

$$18 \times 2 = 36,$$

$$36 \times 2.5 = 90$$

$$90 \times 3 = 270$$

$$? = 270 \times 3.5 = 945$$

Q16 Text Solution:

$$185, 266, 330, 379, 415, \mathbf{440}$$

$$81 \quad 64 \quad 49 \quad 36 \quad 25$$

∴ The answer is 440.

Q17 Text Solution:

$$3 \quad 83 \quad 152 \quad 208 \quad 249 \quad \mathbf{273}$$

$$80 \quad 69 \quad 56 \quad 41 \quad 24$$

$$-11 \quad -13 \quad -15 \quad -17$$

$$? = 273$$

Q18 Text Solution:

The series is difference of difference.

$$15, 28, 48, 82, 137, ?$$

$$13 \quad 20 \quad 34 \quad 55 \quad 83$$

$$7 \quad 14 \quad 21 \quad 28$$

$$? = 137 + 83 = 220$$

Q19 Text Solution:

The pattern of the series –

$$21 \quad 95 \quad 167 \quad 235 \quad 297 \quad 351 \quad 395$$

$$+74 \quad +72 \quad +68 \quad +62 \quad +54 \quad +44$$

$$-2 \quad -4 \quad -6 \quad -8 \quad -10$$

The wrong number in the series is 300.

Q20 Text Solution:

$$9600, 1920, 480, ?, 80, 80$$

the series is, $\div 5, \div 4, \div 3, \div 2, \div 1$

$$? = 480 \div 3$$

$$? = 160$$



Level-2

Q1 Text Solution:

Pattern of the series is as follows :

$$960 - (11)^2 = 839$$

$$839 - (9)^2 = 758$$

$$758 - (7)^2 = 709$$

$$709 - (5)^2 = \mathbf{684}$$

$$684 - (3)^2 = 675$$

$$? = 684$$

$\therefore$ The answer is 684.

Q2 Text Solution:

Here the pattern is,

$$5 = 4 + 1 = 2^2 + 1$$

$$10 = 9 + 1 = 3^2 + 1$$

$$17 = 16 + 1 = 4^2 + 1$$

$$26 = 25 + 1 = 5^2 + 1$$

$$37 = 36 + 1 = 6^2 + 1$$

Now,

$$50 = 7^2 + 1$$

Hence, Option B is Correct

Q3 Text Solution:

$$10 \times 2 = 20$$

$$20 + 2 = 22$$

$$22 \times 2 = 44$$

$$44 + 2 = 46$$

$$46 \times 2 = 92$$

$$92 + 2 = 94$$

Q4 Text Solution:

$$4^2 + 2 = 18$$

$$5^3 + 2 = 127$$

$$6^2 + 2 = 38$$

$$7^3 + 2 = 345$$

$$8^2 + 2 = 66$$

$$9^3 + 2 = 731$$

Q5 Text Solution:

$$181 \times 2 - 1 = 361$$

$$361 \times 2 - 2 = 720$$

$$720 \times 2 - 3 = 1437$$

$$1437 \times 2 - 4 = 2870$$

$$2870 \times 2 - 5 = 5735$$

Q6 Text Solution:

Pattern of the series is as follows :

$$7 \times 2 + 2 = 16$$

$$16 \times 3 - 3 = 45$$

$$45 \times 4 + 4 = 184$$

$$184 \times 5 - 5 = 915$$

$$915 \times 6 + 6 = 5496$$

$$? = 5496$$

Q7 Text Solution:

Pattern of the series is as follows :

$$15 + 6 = 21$$

$$21 + 17 = 38$$

$$38 + 27 = 65$$

$$65 + 36 = 101$$

$$101 + 44 = 145$$

$$\text{Here, } 6 + 11 = 17$$

$$17 + 10 = 27$$

$$27 + 9 = 36$$

$$36 + 8 = 44$$

$$\text{And } 11 - 1 = 10$$

$$10 - 1 = 9$$

$$9 - 1 = 8$$

$$? = 145$$

Q8 Text Solution:

Pattern of the series is as follows :

$$0 = 1^2 - 1$$

$$2 = 2^2 - 2$$

$$8 = 3^2 - 1$$

$$14 = 4^2 - 2$$

$$24 = 5^2 - 1$$

$$34 = 6^2 - 2$$

$$? = 24$$

Q9 Text Solution:

Pattern of the series is as follows :

$$12 \times 1 + 2 = 14$$

$$14 \times 2 + 4 = 32$$

$$32 \times 3 + 6 = 102$$

$$102 \times 4 + 8 = 416$$

$$416 \times 5 + 10 = 2090$$

$$2090 \times 6 + 12 = 12552$$



$$? = 12552$$

Q10 Text Solution:

Pattern of the series is as follows :

$$14 \times 0.5 - 1 = 6 ,$$

$$6 \times 1 - 1 = 5 ,$$

$$5 \times 1.5 - 1 = 6.5 ,$$

$$6.5 \times 2 - 1 = 12 ,$$

$$12 \times 2.5 - 1 = \mathbf{29}$$

$$? = 29$$

Q11 Text Solution:

Pattern of the series is as follows :

$$17 \times 1 - 1 = 16$$

$$16 \times 2 - 2 = 30$$

$$30 \times 3 - 3 = 87$$

$$87 \times 4 - 4 = 344$$

$$344 \times 5 - 5 = 1715$$

$$? = 1715$$

Q12 Text Solution:

Pattern of the series is as follows :

$$24 + (2)^2 = 28$$

$$28 - (3)^2 = 19$$

$$19 + (4)^2 = 35$$

$$35 - (5)^2 = 10$$

$$10 + (6)^2 = 46$$

$$? = 46$$

Q13 Text Solution:

Pattern of the series is as follows :

$$26 \times \frac{1}{2} - 1 = 12$$

$$12 \times 1 - 1 = 11$$

$$11 \times \frac{3}{2} - 1 = 15.5$$

$$15.5 \times 2 - 1 = 30$$

$$30 \times \frac{5}{2} - 1 = 74$$

$$? = 74$$

Q14 Text Solution:

Pattern of the series is as follows :

$$3 \times 2 - 1 = \mathbf{5}$$

$$5 \times 3 - 1 = 14$$

$$14 \times 4 - 1 = 55$$

$$55 \times 5 - 1 = 274$$

$$274 \times 6 - 1 = 1643$$

$$? = 5$$

Q15 Text Solution:

Pattern of the series is as follows :

$$39 = 13 \times 3$$

$$52 = 13 \times 4 (= 1 + 3)$$

$$78 = 13 \times 6 (= 2 + 4)$$

$$117 = 13 \times 9 (= 3 + 6)$$

$$169 = 13 \times 13 (= 4 + 9)$$

$$234 = 13 \times 18 (= 5 + 13)$$

Q16 Text Solution:

$$29 + 1^3 + 1 = 31$$

$$31 + 2^3 + 2 = 41$$

$$41 + 3^3 + 3 = 71$$

$$71 + 4^3 + 4 = 139$$

$$139 + 5^3 + 5 = 269$$

Similarly,

$$84 + 3^3 + 3 = 114$$

$$114 + 4^3 + 4 = 182 \text{ (Ans)}$$

Q17 Text Solution:

$$8 \times 1 - 1^2 = 7$$

$$7 \times 2 - 2^2 = 10$$

$$10 \times 3 - 3^2 = 21$$

$$21 \times 4 - 4^2 = 68$$

$$68 \times 5 - 5^2 = 315$$

Similarly:

$$12 \times 1 - 1^2 = 11$$

$$11 \times 2 - 2^2 = 18$$

$$18 \times 3 - 3^2 = 45$$

$$45 \times 4 - 4^2 = 164$$

$$164 \times 5 - 5^2 = 795$$

So, the value of $m = 795$

Hence, option d.

Q18 Text Solution:

Pattern of series –

$$36 \times 0.5 + 1 = 19$$

$$19 \times 1 + 2 = 21$$

$$21 \times 2 + 3 = 45$$

$$Y = 45 \times 4 + 4 = 184$$

$$184 \times 8 + 5 = 1477$$

$$\text{So, } A + 124 = 2 \times 184 - 48$$

$$A = 320 - 124$$

$$A = 196$$

Q19 Text Solution:

$$9050 - 15^3 = 5675,$$

$$5675 - 13^3 = 3478(x)$$

$$3478 - 11^3 = 2147,$$

$$2147 - 9^3 = 1418 (y)$$

$$1418 - 7^3 = 1075,$$

$$1075 - 5^3 = 950$$

$$\text{Required sum} = 2x + 6\text{th term}$$

$$= 2 \times 3478 + 1075$$

$$= 6956 + 1075 = 8031$$

Q20 Text Solution:

Series is $+2^2 + 3^3 + 4^2 + 5^3 + 6^2$

$$\therefore x = 76 + 5^3 = 76 + 125 = 201$$

$$y = 201 + 6^2 = 201 + 36$$

$$= 237$$

$$(x - y)^2 = (201 - 237)^2$$

$$= (-36)^2$$

$$= 1296$$



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Level-3

Q1. Text Solution:

From the common explanation, we have

$$\frac{a}{c} = \frac{306}{75} = 4.08$$

Hence, Option D is correct.

Common explanation :

Series I :

Consecutive prime numbers:

Series Pattern	Given Series	
$17 \times 17 + 17 = 306$	306	a
$13 \times 13 + 13 = 182$	182	
$11 \times 11 + 11 = 132$	132	
$7 \times 7 + 7 = 56$	56	
$5 \times 5 + 5 = 30$	30	
$3 \times 3 + 3 = 12$	12	
$2 \times 2 + 2 = 6$	6	b

Series II :

6th term of the series = $b = a - 4c$

$$306 - 4c = 6$$

$$c = 75$$

$$a - 5c = d$$

$$306 - 5 \times 75 = -69 = d$$

Q2. Text Solution:

From the common explanation, we have

$$b - d = 6 - (-69) = 75$$

$$\text{option A} = \frac{a}{4} = \frac{306}{4} = 76.50 \text{ (Eliminate)}$$

Option B = $b = 6$ (Eliminate)

Option C = $c = 75$ (Satisfies)

Hence, Option C is correct.

Common explanation :

Series I :

Consecutive prime numbers:

Series Pattern	Given Series	
$17 \times 17 + 17 = 306$	306	a
$13 \times 13 + 13 = 182$	182	
$11 \times 11 + 11 = 132$	132	
$7 \times 7 + 7 = 56$	56	
$5 \times 5 + 5 = 30$	30	
$3 \times 3 + 3 = 12$	12	
$2 \times 2 + 2 = 6$	6	b

Series II :

6th term of the series = $b = a - 4c$

$$306 - 4c = 6$$

$$c = 75$$

$$a - 5c = d$$

$$306 - 5 \times 75 = -69 = d$$

Q3. Text Solution:

From common explanation, we have

$$\frac{d}{a} = \frac{39270}{12} = 3272.50$$

Hence, Option B is correct.



Common explanation :

Series 1 :

Series Pattern	Given Series	
12	12	a
$12 \times 12 + 10 = 154$	154	
$154 \times 10 + 8 = 1548$	1548	
$1548 \times 8 + 6 = 12390$	12390	
$12390 \times 6 + 4 = 74344$	74344	
$74344 \times 4 + 2 = 297378$	297378	

Series 2 :

Series Pattern	Given Series	
2	2	b
$2 \times 14 + 12 = 40$	40	c
$40 \times 12 + 10 = 490$	490	
$490 \times 10 + 8 = 4908$	4908	
$4908 \times 8 + 6 = 39270$	39270	d
$39270 \times 6 + 4 = 235624$	235624	

Q4. Text Solution:

From common explanation, we have

$$\frac{a}{3}, 4b, \frac{c}{2.50}, 4b^3, e$$

$$\frac{12}{3}, 4 \times 2, \frac{40}{2.50}, 4 \times 8, e$$

$$4, 8, 16, 32, e$$

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

$$32 \times 2 = 64$$

So the answer = 64

Hence, Option B is correct.

Common explanation :

Series 1 :

Series Pattern	Given Series	
12	12	a
$12 \times 12 + 10 = 154$	154	
$154 \times 10 + 8 = 1548$	1548	
$1548 \times 8 + 6 = 12390$	12390	
$12390 \times 6 + 4 = 74344$	74344	
$74344 \times 4 + 2 = 297378$	297378	

Series 2 :

Series Pattern	Given Series	
2	2	b
$2 \times 14 + 12 = 40$	40	c
$40 \times 12 + 10 = 490$	490	
$490 \times 10 + 8 = 4908$	4908	
$4908 \times 8 + 6 = 39270$	39270	d
$39270 \times 6 + 4 = 235624$	235624	

Q5 Text Solution:

3rd least prime number is 5

$$\text{So, value of } Q = 5^3 + 1 = 126$$

$$\text{And value of } P = 126 \times \frac{5}{21} = 30$$

Logic in the series:

$$14 \times 2 + 2 = 30$$

$$30 \times 2 + 2 = 62$$

$$62 \times 2 + 2 = 126$$

$$126 \times 2 + 2 = 254$$

$$\text{And next term of the series} = 254 \times 2 + 2 = 510$$

Q6 Text Solution:

Logic in the given series:

$$8 \times 2.5 = 20$$

$$20 \times 5 = 100$$

$$100 \times 7.5 = 750$$



$$750 \times 10 = 7500$$

$$7500 \times 12.5 = 93750$$

Hence, the wrong term in the given series = 150

Now,

$$1\text{st term of the new series} = 150$$

$$2\text{nd term of the new series} = 150 + 64 = 214$$

$$3\text{rd term of the new series} = 214 - 49 = 165$$

$$4\text{th term of the new series} = 165 + 36 = 201$$

$$5\text{th term of the new series} = 201 - 25 = 176$$

$$\text{So, 5th term of the new series} = 176$$

Q7 Text Solution:

Logic in the series:

$$18 \times 2 + 1 = 37$$

$$37 \times 2 + 1 = 75$$

$$75 \times 2 + 1 = 151$$

$$151 \times 2 + 1 = 303$$

$$303 \times 2 + 1 = 607$$

$$\text{So, } X = 303 \text{ and } Y = 607$$

From A:

$$X = 303, Y = 607$$

X is a multiple of 3 but Y is not a multiple of 3.

So, A is not true.

From B:

$$X = \frac{Y-1}{2}$$

$$303 = \frac{607-1}{2}$$

$$303 = 303$$

So, B is true.

From C:

$$Y = 607$$

Y is a prime number.

So, C is true.

Hence, only B and C are true.

Q8 Text Solution:

Logic in the given series:

$$325 - 5^3 = 200$$

$$200 - 4^3 = 136$$

$$136 - 3^3 = 109$$

$$109 - 2^3 = 101$$

$$101 - 1^3 = 100$$

$$\text{So, the value of } P = 200$$

Now,

$$1\text{st term of the new series} = 200$$

$$2\text{nd term of the new series} = 200 - 5^3 = 75$$

$$3\text{rd term of the new series} = 75 - 4^3 = 11$$

$$4\text{th term of the new series} = 11 - 3^3 = -16$$

Q9 Text Solution:

Logic in the given series:

$$8 \times 1 + 1 = 9$$

$$9 \times 2 + 2 = 20$$

$$20 \times 3 + 3 = 63$$

$$63 \times 4 + 4 = 256$$

$$256 \times 5 + 5 = 1285$$

$$\text{So, the missing term in the given series} = P = 9$$

Now,

$$1\text{st term of the new series} = 9 - 3 = 6$$

$$2\text{nd term of the new series} = 6 \times 1 + 1 = 7$$

$$3\text{rd term of the new series} = 7 \times 2 + 2 = 16$$

$$4\text{th term of the new series} = 16 \times 3 + 3 = 51$$

$$5\text{th term of the new series} = 51 \times 4 + 4 = 208$$

$$\text{So, the value of 'n' = 5}$$

Q10 Text Solution:

Logic in the given series:

$$8 \times 0.5 = 4$$

$$4 \times 1 = 4$$

$$4 \times 2 = 8$$

$$8 \times 4 = 32$$

$$32 \times 8 = 256$$

$$\text{So, the missing term of the given series} = 32$$

Now,

$$1\text{st term of the new series} = 32$$

$$2\text{nd term of the new series} = 32 \times 0.25 = 8$$

$$3\text{rd term of the new series} = 8 \times 0.5 = 4$$

$$4\text{th term of the new series} = 4 \times 1 = 4$$

$$5\text{th term of the new series} = 4 \times 2 = 8$$

$$6\text{th term of the new series} = 8 \times 4 = 32$$

